

Medical Terminology – BBAHM 201

Complete Summary Notes | All 5 Modules Covered

Subject: Medical Terminology | **Code:** BBAHM 201 (Major) | **Credits:** 5 (4T+1P)

University: Maulana Abul Kalam Azad University of Technology, West Bengal (MAKAUT)

Program: BBA in Hospital Management | **Effective From:** 2023-24 Academic Sessions

Semester: 2nd Semester | **Full Marks:** 70

Table of Contents

1. **Module 1** — Basics of Human Body, Body Cavities, Body Planes & Basic Examination; Cell & Cell Division
 2. **Module 2** — Introduction to Medical Terminology; Word Formation; Prefixes, Suffixes, Root Words; Body Part Terms
 3. **Module 3** — Human System Diseases & Procedures: Cardiovascular, Digestive, Nervous, Urinary, Endocrine, Reproductive, Respiratory
 4. **Module 4** — Prescription Reading; Latin Terms & Abbreviations; Dosage Forms; Measurement Units
 5. **Module 5** — Diagnostic Procedures: Fundamentals + Cardiology, Gastroenterology, Reproductive, Pulmonology, Orthopaedic Departments
-

Basics of Human Body, Body Cavities, Body Planes & Basic Examination

1.1 Levels of Organization of the Human Body

The human body is organized from simplest to most complex in a hierarchical manner:

Level	Description	Example
Chemical	Atoms and molecules	Water (H ₂ O), DNA, proteins
Cellular	Basic structural unit of life	Neuron, Red Blood Cell
Tissue	Group of similar cells performing same function	Muscle tissue, Nervous tissue
Organ	Two or more tissue types working together	Heart, Liver, Kidney
Organ System	Group of organs with related functions	Cardiovascular, Digestive
Organism	All systems together forming a living being	Human being

1.2 Body Cavities

Body cavities are spaces within the body that contain and protect internal organs.

Dorsal Cavity

- **Cranial cavity** — encloses the brain
- **Spinal (vertebral) cavity** — encloses the spinal cord

Ventral Cavity

- **Thoracic cavity** — heart, lungs, trachea
- **Abdominopelvic cavity** — digestive organs, kidneys, bladder, reproductive organs

Abdominopelvic 9 regions: Right Hypochondriac | Epigastric | Left Hypochondriac | Right Lumbar | Umbilical | Left Lumbar | Right Iliac | Hypogastric | Left Iliac

4 Quadrants: RUQ (liver, gallbladder) | LUQ (stomach, spleen) | RLQ (appendix, cecum) | LLQ (sigmoid colon)

1.3 Body Planes (Anatomical Planes)

Plane	Also Called	Division
Sagittal	Lateral plane	Left and right halves; Midsagittal = equal halves
Frontal	Coronal plane	Anterior (front) and posterior (back)
Transverse	Horizontal / Axial plane	Superior (upper) and inferior (lower)

1.4 Directional Terms

Term	Meaning	Term	Meaning
Superior / Cranial	Toward the head	Inferior / Caudal	Toward the feet
Anterior / Ventral	Front of body	Posterior / Dorsal	Back of body
Medial	Toward midline	Lateral	Away from midline
Proximal	Closer to trunk	Distal	Farther from trunk
Superficial	Near body surface	Deep	Away from surface

1.5 Basic Physical Examination

1. Inspection

Visual assessment — skin, posture, breathing

2. Palpation

Feeling with hands — texture, tenderness, mass

3. Percussion

Tapping — dull (solid/fluid), resonant (air)

4. Auscultation

Listening via stethoscope — heart, lungs, bowel

Normal Vital Signs: Temperature 98.6°F / 37°C | Pulse 60–100 bpm | RR 12–20/min | BP 120/80 mmHg | SpO₂ 95–100%

Cell & Cell Division

2.1 Cell Structure — Key Organelles

Organelle	Function
Cell membrane	Selective barrier; controls entry/exit
Nucleus	Control centre; contains DNA
Mitochondria	ATP production — powerhouse of the cell
Ribosome	Protein synthesis
Rough ER	Protein processing
Smooth ER	Lipid synthesis, detoxification
Golgi apparatus	Packages and ships proteins — post office of cell
Lysosome	Digests waste — suicide bag
Centrioles	Spindle formation during cell division

2.2 Cell Division

Mitosis

- 2 identical daughter cells (diploid, 2n=46)
- Purpose: Growth, repair
- Phases: PMAT

Meiosis

- 4 genetically unique cells (haploid, n=23)
- Purpose: Gamete production
- Two rounds: Meiosis I and II

Mitosis Phase	Key Events
Prophase	Chromosomes condense; nuclear envelope breaks; spindle forms
Metaphase	Chromosomes align at equator (metaphase plate)
Anaphase	Sister chromatids separate to opposite poles
Telophase + Cytokinesis	Nuclear envelopes reform; cytoplasm divides into 2 cells

Clinical relevance: Uncontrolled mitosis → Cancer. Meiotic non-disjunction → Down syndrome (Trisomy 21), Turner (45,X), Klinefelter (47,XXY).

Introduction to Medical Terminology & Word Formation

2.1 What is Medical Terminology?

Medical terminology is a standardized language used by healthcare professionals, primarily derived from **Greek and Latin**, to accurately communicate about the human body, diseases, procedures, and treatments with universal precision.

2.2 Word Formation — Components

Component	Definition	Position	Example
Root Word	Core meaning	Middle	cardi (heart), hepat (liver)
Prefix	Modifies root; adds detail	Beginning	tachy- (fast), brady- (slow)
Suffix	Indicates condition/procedure	End	-itis (inflammation), -ectomy (removal)
Combining vowel	Links parts (usually 'o')	Between parts	cardiology (cardi-o-logy)

Rule: Combining vowel used between root words or before consonant-beginning suffixes. Dropped before vowel-beginning suffixes. Example: gastr/o + itis = gastritis (dropped).

2.3 Greek & Latin Prepositional Prefixes

Prefix	Meaning	Example	Prefix	Meaning	Example
anti-	Against	Antibiotic	hypo-	Below normal	Hypoglycemia
peri-	Around	Pericardium	hyper-	Above normal	Hypertension
epi-	Upon, over	Epidermis	endo-	Within	Endoscopy
intra-	Within	Intravenous	sub-	Under	Subcutaneous
supra-	Above	Suprarenal	trans-	Across	Transdermal
pre-	Before	Prenatal	post-	After	Postoperative
inter-	Between	Intercostal	exo-	Outside	Exocrine

2.4 Key Prefixes — Number, Color, State

Prefix	Meaning	Example	Color Prefix	Color	Example
mono-/uni-	One	Mononuclear	erythr/o-	Red	Erythrocyte (RBC)
bi-/di-	Two	Bilateral	leuk/o-	White	Leukocyte (WBC)
poly-/multi-	Many	Polyneuropathy	cyan/o-	Blue	Cyanosis
micro-	Small	Microcephaly	melan/o-	Black/dark	Melanoma
macro-/mega-	Large	Megacolon	xanth/o-	Yellow	Xanthoma
a-/an-	Without	Apnea, Anemia	alb/o-	White	Albinism
dys-	Difficult/painful	Dyspnea			
brady-	Slow	Bradycardia			
tachy-	Fast	Tachycardia			
hemi-	Half	Hemiplegia			

2.5 Key Suffixes

Suffix	Meaning	Example	Suffix	Meaning	Example
-algia/-dynia	Pain	Neuralgia	-ectomy	Surgical removal	Appendectomy

-itis	Inflammation	Hepatitis	-otomy	Surgical incision	Laparotomy
-osis	Abnormal condition	Fibrosis	-ostomy	Surgical opening	Colostomy
-emia	Blood condition	Anaemia	-plasty	Surgical repair	Angioplasty
-pathy	Disease	Neuropathy	-scopy	Visual exam	Colonoscopy
-rrhea	Flow/discharge	Diarrhoea	-graphy	Recording process	Radiography
-plegia	Paralysis	Hemiplegia	-gram	Record/image	ECG
-megaly	Enlargement	Hepatomegaly	-logy	Study of	Cardiology
-stenosis	Narrowing	Aortic stenosis	-centesis	Surgical puncture	Amniocentesis

2.6 Common Root Words — Body Systems

Root	Meaning	Root	Meaning
cardi/o	Heart	oste/o	Bone
angi/o	Blood vessel	arthr/o	Joint
hem/o	Blood	my/o	Muscle
hepat/o	Liver	hyster/o	Uterus
gastr/o	Stomach	rhin/o	Nose
nephr/o	Kidney	ophthalm/o	Eye
pulmon/o	Lung	ot/o	Ear
neur/o	Nerve	adren/o	Adrenal gland
encephal/o	Brain	cyst/o	Bladder
dermat/o	Skin	col/o	Colon

Human System Diseases & Procedures

♥ Cardiovascular System

Disease	Definition & Key Features
IHD	Ischaemic Heart Disease — reduced coronary blood flow; umbrella term (includes angina, MI)
Angina Pectoris	Chest pain due to insufficient myocardial O ₂ ; stable (exertion) vs. unstable (rest)
Heart Valve Disease	Valvular stenosis (narrowing) or regurgitation (leaking) — mitral, aortic, tricuspid, pulmonary
CHD	Congenital Heart Disease — structural defect at birth (VSD, ASD, PDA, Tetralogy of Fallot)
Anaemia	Reduced Hb/RBC → low O ₂ carrying capacity; Iron deficiency (most common), pernicious, aplastic
Thalassaemia	Inherited defect in globin chain synthesis → chronic haemolytic anaemia; alpha & beta types
Haemophilia	Inherited clotting factor deficiency; Type A (Factor VIII), Type B (Factor IX); X-linked recessive

Cardiovascular Procedures:

- **Angioplasty (PTCA)** — balloon catheter inflates in blocked coronary artery; stent placed
- **CABG** — bypass blocked coronary arteries using saphenous vein or IMA graft; open-heart surgery
- **CPR** — 30 chest compressions : 2 breaths; 100–120/min; restores circulation in cardiac arrest
- **Defibrillation** — electric shock to restore rhythm in VF/VT; AED available for public use
- **Pacemaker** — electrical device implanted for bradycardia; regulates slow heart rate
- **Heart Transplantation** — donor heart replacement for end-stage heart failure; lifelong immunosuppression

🍷 Digestive System

Disease	Definition & Key Features
Peptic Ulcer	Open sore in stomach/duodenum lining; H. pylori or NSAIDs; burning epigastric pain
GERD	Gastro-Oesophageal Reflux Disease; backflow of acid; heartburn, regurgitation; weak LES
Dyspepsia	Indigestion; epigastric pain/bloating/nausea; functional or organic cause
Jaundice	Excess bilirubin → yellow skin/eyes; pre-hepatic / hepatic / post-hepatic classification
Gallstone (Cholelithiasis)	Hardened deposits in gallbladder; RUQ biliary colic; can cause cholecystitis
Hepatitis	Liver inflammation; viral (A/B/C/D/E), alcoholic, autoimmune; jaundice, elevated liver enzymes

Digestive Procedures:

- **Appendectomy** — removal of appendix; acute appendicitis; open or laparoscopic
- **Whipple Procedure** — pancreaticoduodenectomy; head of pancreas + duodenum + gallbladder; pancreatic cancer
- **Cholecystectomy** — gallbladder removal; mostly laparoscopic; for symptomatic gallstones
- **Hiatal Hernia Repair** — correction of stomach herniation through diaphragm
- **Nephrectomy** — kidney removal; cancer, trauma, donor transplant
- **Nissen Fundoplication** — wraps stomach fundus around lower oesophagus; strengthens LES; treats severe GERD

Nervous System

Disease	Definition & Key Features
Stroke (CVA)	Sudden brain infarction; ischaemic (clot, 80%) or haemorrhagic; FAST = Face, Arm, Speech, Time
Quadriplegia	Paralysis of all 4 limbs; cervical spinal cord injury (C1–C7)
Bell's Palsy	Unilateral facial nerve (CN VII) paralysis; drooping of one side of face; usually reversible
Parkinson's Disease	Dopamine depletion; TRAP = Tremor, Rigidity, Akinesia, Postural instability
Alzheimer's Disease	Most common dementia; amyloid plaques + tau tangles; progressive memory loss; no cure

Nervous System Procedures:

- **Awake Brain Surgery (Awake Craniotomy)** — patient conscious; real-time monitoring near speech/motor cortex
- **Brain Aneurysm Surgery** — Clipping (surgical) or Coiling (endovascular); prevents subarachnoid haemorrhage
- **Epilepsy Surgery** — temporal lobectomy or corpus callosotomy; drug-resistant epilepsy
- **Locomotor Training** — treadmill + body-weight support; promotes neuroplasticity in SCI
- **Trigeminal Neuralgia Surgery** — Microvascular decompression (MVD); relieves pressure on CN V
- **Deep Brain Stimulation (DBS)** — electrode implant in subthalamic nucleus; treats Parkinson's, tremor

Urinary System

Disease	Key Features
Dialysis	Artificial waste removal; haemodialysis (machine) vs. peritoneal (abdomen); used in CKD/ESRD
Nephritis	Kidney inflammation; glomerulonephritis; haematuria, proteinuria, oedema, hypertension
BPH	Benign Prostatic Hyperplasia; men >50; obstructs urethra; frequency, urgency, nocturia
Hydronephrosis	Kidney swelling from obstructed urine flow; stones, stricture, BPH; risk of renal damage

Endocrine System

Disease	Key Features
Diabetes Mellitus	Type 1 (insulin deficiency) / Type 2 (insulin resistance); polydipsia, polyuria, polyphagia
Diabetic Foot	Neuropathy + poor circulation → ulcers, infection; high amputation risk
Gangrene	Tissue death; dry (vascular), wet (infected), gas (Clostridium); major diabetic complication
Hyposecretion diseases	Hypothyroidism, Addison's disease, Hypopituitarism, Hypoparathyroidism
Hypersecretion diseases	Hyperthyroidism/Graves', Cushing's (excess cortisol), Acromegaly (excess GH adult), Gigantism (excess GH child)

Reproductive System

Disease	Key Features
Infertility	Failure to conceive after 12 months; female (ovulatory, tubal) or male (sperm) factors
Endometriosis	Endometrial tissue outside uterus; pelvic pain, dysmenorrhoea, infertility
PCOS	Polycystic Ovary Syndrome; hormonal; irregular periods, excess androgens, insulin resistance
PCOD	Polycystic Ovarian Disease; structural (multiple cysts); milder than PCOS
Uterine Fibroids	Benign smooth muscle tumours (leiomyomas); heavy periods, pelvic pressure; rarely cancerous

Syphilis	Treponema pallidum STI; stages: primary chancre → secondary rash → latent → tertiary; treated with penicillin
Prostatitis	Prostate inflammation; bacterial or non-bacterial; pelvic pain, dysuria
Erectile Dysfunction	Inability to maintain erection; vascular/neurological/hormonal/psychological; PDE5 inhibitors
Epididymitis	Epididymis inflammation; bacterial STI or UTI; scrotal pain/swelling; antibiotics

Reproductive Procedures:

- **Hysterectomy** — uterus removal; total/partial/radical; for fibroids, endometriosis, cancer, prolapse
- **Laparotomy** — abdominal cavity incision; exploratory surgery, ovarian cysts, obstetric emergencies

 Respiratory System

Disease	Key Features
Asthma	Reversible bronchospasm; allergens/exercise triggers; wheeze, cough, dyspnoea; salbutamol + ICS
Pneumonia	Alveolar infection; S. pneumoniae (most common); fever, cough, pleuritic pain; CXR consolidation
Tuberculosis (TB)	Mycobacterium tuberculosis; cough, haemoptysis, night sweats; RIPE therapy × 6 months
COPD	Progressive airflow obstruction; smoking main cause; chronic bronchitis + emphysema; FEV ₁ /FVC < 0.70
Emphysema	Alveolar destruction; loss of elastic recoil; barrel chest; component of COPD; smoking

Key Distinctions: PCOS (hormonal syndrome) vs. PCOD (structural, milder) | Quadriplegia (cervical) vs. Paraplegia (thoracic+) | Haemodialysis (3×/week machine) vs. Peritoneal (daily, home) | Dry gangrene (vascular, no infection) vs. Wet (infected, spreads rapidly)

Prescription Reading & Latin Abbreviations

4.1 Parts of a Prescription

Part	Description
Superscription	Rx symbol + patient details (name, age, date)
Inscription	Drug name and strength
Subscription	Dispensing instructions to pharmacist (quantity)
Signa (Sig)	Patient directions — dose, route, frequency, duration
Refill	Number of refills allowed
Prescriber	Signature, reg. number, stamp

4.2 Frequency Abbreviations

Abbreviation	Latin	Meaning	Abbreviation	Latin	Meaning
OD / QD	Omni die	Once daily	HS	Hora somni	At bedtime
BD / BID	Bis in die	Twice a day	SOS / PRN	Si opus sit	If needed
TDS / TID	Ter in die	Three times a day	STAT	Statim	Immediately
QID / QDS	Quater in die	Four times a day	NPO / NBM	Nil per os	Nothing by mouth
AC	Ante cibum	Before meals	PC	Post cibum	After meals
CC	Cum cibum	With meals	WNL	—	Within Normal Limits

4.3 Route of Administration

Abbrev.	Meaning	Abbrev.	Meaning
PO	By mouth (per os)	IV	Intravenous
IM	Intramuscular	SC / SQ	Subcutaneous
SL	Sublingual	INH	Inhaled
PR	Per rectum	TOP	Topical
OD (eye)	Right eye	OS (eye)	Left eye
OU	Both eyes	ID	Intradermal

4.4 Dosage Forms

Abbreviation	Latin	Form	Abbreviation	Latin	Form
Tab	Tabella	Tablet	Inj	Injectio	Injection
Cap	Capsula	Capsule	Oint/Ung	Unguentum	Ointment
Syr	Syrupus	Syrup	Gtt	Guttae	Drops
Supp	Suppositorium	Suppository	Susp	Suspensio	Suspension

4.5 Clinical Abbreviations (Commonly Used)

Abbrev.	Full Form	Abbrev.	Full Form
Dx	Diagnosis	CBC	Complete Blood Count
Hx	History	LFT	Liver Function Test
Sx	Symptoms	KFT/RFT	Kidney/Renal Function Test
Tx	Treatment	FBS	Fasting Blood Sugar

k/c/o	Known case of	ECG	Electrocardiogram
h/o	History of	USG	Ultrasonography
c/o	Complaining of	CXR	Chest X-Ray
R/O	Rule Out	OPD	Outpatient Department
NAD	No Abnormality Detected	ICU	Intensive Care Unit
NKDA	No Known Drug Allergies	MLC	Medico-Legal Case

Diagnostic Procedures

5.1 Fundamental Imaging Modalities

Modality	Principle	Radiation	Clinical Use
X-Ray	X-rays absorbed by dense tissue (bone), pass through soft tissue; 2D shadow image	Yes (low)	Fractures, CXR (pneumonia, TB), joints
USG	High-frequency sound waves reflect off tissues; echoes → image	None	Abdomen, obstetrics, thyroid, soft tissue; safe in pregnancy
CT Scan	Multiple X-ray beams rotate; computer reconstructs cross-sectional images	Yes (moderate-high)	Head trauma, stroke, tumours, internal injuries
MRI	Magnetic field + radiofrequency pulses; hydrogen proton realignment signals → image; no radiation	None	Brain, spine, joints, soft tissue; superior to CT for soft tissue
PET Scan	FDG radiotracer → cancer cells absorb more glucose → positron emission → gamma rays detected	Yes (radiotracer)	Cancer staging, Alzheimer's, cardiac viability; combined with CT = PET-CT
Biopsy	Tissue sample removal for pathological examination	None	Definitive cancer diagnosis; types: excisional, incisional, core needle, FNAC
FNAC	Fine needle (22–25G) aspirates cells for cytology	None	Thyroid nodules, breast lumps, lymph nodes; minimally invasive, outpatient

♥ Cardiology Department

Procedure	Description & Clinical Use
ECG	Electrocardiogram — records heart electrical activity; 12-lead standard; detects MI (STEMI = ST elevation), arrhythmias, heart blocks, hypertrophy; first-line cardiac test
Echocardiography	Cardiac ultrasound; assesses chambers, valves, ejection fraction, wall motion; types: 2D, 3D, Doppler, TOE; for heart failure, valve disease, CHD
Coronary Angiography	Contrast dye via catheter (femoral/radial) into coronary arteries; X-ray imaging; gold standard for CAD diagnosis; done in Cath Lab
Doppler USG	Measures blood flow direction and speed; colour Doppler maps flow; detects DVT, carotid stenosis, peripheral arterial disease
64-Slice CT	Non-invasive coronary CT angiography; detects hard and soft plaques; alternative to invasive angiography for low-moderate risk patients
Cardiac Catheterization	Catheter into heart chambers; measures pressures + O ₂ levels; combined with angiography; for hemodynamic assessment + PCI interventions
Holter Monitor	Portable ECG worn 24–48 hours; detects intermittent arrhythmias (paroxysmal AF, SVT) missed on standard ECG

🏠 Gastroenterology Department

Procedure	Description & Clinical Use
Colonoscopy	Camera through rectum; visualizes entire colon + terminal ileum; CRC screening, IBD, polyps, biopsy + polypectomy
ERCP	Endoscopic Retrograde Cholangiopancreatography — endoscope to duodenum; contrast into bile/pancreatic ducts; treats bile duct stones (sphincterotomy), stenting; diagnostic + therapeutic
EGD (Upper GI Endoscopy)	Esophagogastroduodenoscopy — scope through mouth; visualizes oesophagus, stomach, duodenum; peptic ulcers, GERD, gastric cancer, H. pylori biopsy
Sigmoidoscopy	Short scope examining sigmoid colon + rectum; rectal bleeding, IBD; faster than colonoscopy; less preparation required

Esophageal Manometry	Pressure catheter in oesophagus; measures LES pressure + contractions; diagnoses achalasia, GERD, oesophageal spasm
Capsule Endoscopy	Swallowed pill-camera; images entire GI tract including small intestine; for obscure GI bleeding, Crohn's, small bowel tumours
MRCP	Magnetic Resonance Cholangiopancreatography — non-invasive MRI of biliary + pancreatic ducts; detects stones, strictures; preferred over diagnostic ERCP

 Reproductive Department

Procedure	Description & Clinical Use
Colposcopy	Magnified examination of cervix after abnormal Pap smear; identifies dysplastic lesions; directed biopsy; cervical cancer screening
HSG	Hysterosalpingography — X-ray of uterus + fallopian tubes with contrast; assesses tubal patency; infertility workup
Scrotal USG	Ultrasound of testes, epididymis; detects testicular torsion, epididymitis, varicocele, testicular cancer; first-line for acute scrotal pain

 Pulmonology Department

Procedure	Description & Clinical Use
Nuclear Lung Scanning (V/Q Scan)	Inhaled + IV radiotracer; gamma camera images; V/Q mismatch = pulmonary embolism; used when CTPA contraindicated (contrast allergy, pregnancy)
Pulmonary Artery Angiography	Catheter into pulmonary artery; contrast injection; X-ray imaging of pulmonary vasculature; gold standard for PE (now replaced by CTPA); essential pre-op for CTEPH surgery

 Orthopaedic Department

Procedure	Description & Clinical Use
Bone Densitometry (DEXA)	Dual-energy X-ray measures Bone Mineral Density (BMD); T-score: Normal ≥ -1.0 Osteopenia: -1.0 to -2.5 Osteoporosis ≤ -2.5 ; post-menopausal women screening
Arthroscopy	Arthroscope inserted into joint through small incision; visualizes cartilage, ligaments, menisci; diagnostic + therapeutic (ACL repair, meniscectomy); most common: knee

Quick Reference — Modality Comparison:			
Modality	Radiation	Invasive	Best For
X-Ray	Low	No	Bones, chest
USG	None	No	Abdomen, obstetrics
CT Scan	Moderate-high	No	Trauma, tumours, head
MRI	None	No	Brain, spine, soft tissue
PET Scan	Radiotracer	Injection only	Cancer, metabolism
Biopsy/FNAC	None	Yes (needle)	Tissue/cancer diagnosis
Angiography	Yes	Yes (catheter)	Coronary/pulmonary vessels